



QZhou-Embedding Technical Report

Peng Yu, En Xu, Bin Chen, Haibiao Chen, Yinfei Xu

Kingsoft AI*

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Abstract

We present QZhou-Embedding, a general-purpose contextual text embedding model with exceptional text representation capabilities. Built upon the Qwen2.5-7B-Instruct foundation model, we designed a unified multi-task framework comprising specialized data transformation and training strategies. The data transformation scheme enables the incorporation of more diverse textual training datasets, while the task-specific training strategies enhance model learning efficiency. We developed a data synthesis pipeline leveraging LLM API, incorporating techniques such as Paraphrasing, Augmentation, and Hard negative example generation to improve the semantic richness and sample difficulty of the training set. Additionally, we employ a two-stage training strategy, comprising initial retrieval-focused pretraining followed by full-task fine-tuning, enabling the embedding model to extend its capabilities based on robust retrieval performance. Our model achieves state-of-the-art results on the MTEB and CMTEB benchmarks, ranking first on both leaderboards(August 27, 2025), simultaneously achieves state-of-the-art performance on tasks including Reranking, Clustering, etc. Our findings demonstrate that higher-quality, more diverse data is crucial for advancing retrieval model performance, and that leveraging LLMs' generative capabilities can further optimize data quality for embedding model breakthroughs. Our model weights are released on HuggingFace¹ under Apache 2.0 license. For reproducibility, we provide evaluation code and instructions on GitHub².

1 Introduction

Text embedding models, which transform natural language text into mathematical vector representations, play an indispensable role in text mining, question-answering systems, recommendation systems, and retrieval-augmented generation. Recently, LLM-based agent technology has experienced rapid development and widespread adoption, embedding models, which transform textual or multimodal data into vector representations for knowledge base construction, have significantly enhanced agent systems

*<https://kingsoft.com/>

¹<https://huggingface.co/Kingsoft-LLM/QZhou-Embedding>

²<https://github.com/Kingsoft-LLM/QZhou-Embedding>

in terms of real-time performance, long-term memory, data privacy preservation, and knowledge integration capabilities. With the continuous advancement of neural networks and deep learning, text embeddings have evolved from early sparse representations (e.g., BM25[1]) to dense representations based on fine-tuned deep networks such as BERT[2] and T5[3], leading to significant performance improvements[4][5][6][7][8]. In 2022, the rise of large language models (LLMs), exemplified by ChatGPT[9], ushered in a new era of text embeddings based on LLM representations, including models like text-embedding-3-large and RepLLaMA[10]. Recent research on optimizing text embedding models has explored diverse perspectives and focal points. For instance, to address the limitation of decoder-only architectures—where causal attention mechanisms restrict token embeddings to unidirectional semantic capture—several approaches have been proposed: Echo Embedding[11] employs input repetition and instruction design to enable preceding tokens to capture subsequent token semantics. LLM2Vec[12] modifies attention to bi-directional mechanism to remove backward dependency constraints. Conan-Embedding-v2[13] proposes a novel soft masking mechanism combined with dynamic rank reduction. Another widely adopted approach is knowledge distillation, where text embeddings are treated as the "signal states" representing textual semantics. By distilling knowledge from high-performing teacher models to student models, the objective is to optimize the embedding performance. For instance, Jasper[14] employs a multi-stage knowledge distillation framework, combining with multiple carefully designed loss functions and finally achieving superior results. Debater[16] proposes a step-by-step thinking mechanism for embedding generation, iteratively optimizing document representations through continuous COT. Distillation is applied to constrain the final token representation to learn the optimal semantic states from these thinking steps. Additionally, hard negative sampling has emerged as a crucial research direction in text embedding models, serving as a pivotal technique for model optimization. ANCE[18] identified that conventional dense retrieval training leads to diminishing gradient norms during optimization. Thus they developed an asynchronous Approximate Nearest Neighbor (ANN) indexing mechanism that periodically refreshes the negative sample pool using the current model parameters, thereby ensuring the maintenance of up-to-date and optimally challenging negative samples. Both Conan-Embedding[24] and its v2 version incorporated similar dynamic hard negative sampling techniques to enhance model performance. NV-Embed[19] implemented an alternative approach by leveraging their previously developed NV-Retriever's[20] positive-aware negative mining strategy, including TopK-MarginPos and TopKPercPos filtering mechanisms.

In this work, we present QZhou-Embedding, built upon the powerful Qwen2.5-7B-Instruct[21] model, which pushes the boundaries of text embedding capabilities. To enhance the model's semantic understanding, we designed a unified multi-task learning framework that not only accommodates more diverse training data but also bring efficient learning across three key tasks: retrieval, natural language inference (NLI), and classification. Our framework comprises two core components: 1. Data Transformation: We carefully adapt data formats to the specific requirements of retrieval, NLI, and classification tasks, enabling effective feature extraction from heterogeneous data sources, significantly benefiting retrieval model training. 2. Training Strategy: We designed specialized loss functions based on each task's characteristics, optimizing model training efficiency. To further improve the robustness and generalization of vec-

tor representation, we propose a data synthesis method by employing three techniques to address data scarcity: Paraphrasing & Data augmentation for limited datasets and Hard negative generation for negative sample enrichment. Building upon prior work, we designed a strategy named "Data Grouping Strategy", enabling batch sampling within single datasets, inadvertently increasing training difficulty through in-batch negative sampling from the same distribution. For model training, we used a two-phase training approach, through the first-stage retrieval training and second-stage full-capability training, our model acquires a solid foundation of retrieval capabilities, while effectively extending to multiple capability dimensions. Our model achieved state-of-the-art average scores on CMTEB[22] and MTEB[23] benchmarks, ranking first overall on both CMTEB and MTEB leaderboards, demonstrating the effectiveness of our approach. The contributions of our work are summarized as follows:

- We propose a unified multi-task learning framework that systematically coordinates both data processing and training pipelines, enhancing diversity in datasets and efficiency in model training ;
- We develop advanced data synthesis techniques powered by LLM, including Paraphrasing, Data augmentation, and Hard negative generation. These methods significantly enhance the quality of training corpora, thereby improving model's robustness and generalization capabilities;
- We employ a two-stage training paradigm: Stage 1 focuses exclusively on retrieval capability building, establishing strong foundational retrieval performance; and stage 2 implements balanced training with controlled retrieval/non-retrieval task ratios, achieving superior performance on classification (CLS), pair classification (PairCLS), and semantic textual similarity (STS) tasks while maintaining retrieval effectiveness;
- Our model achieves state-of-the-art performance on both MTEB and CMTEB benchmarks, which validates the effectiveness of our proposed methods.

2 Related Works

2.1 Text Embedding Models

Text vector representation is a fundamental research area in natural language processing (NLP) and serves as the cornerstone for language understanding. Early approaches relied on sparse vector representations, such as TF-IDF[25], BM25[26], and LSA[27]. With the advent of pretrained language models, dense contextualized representations based on architectures like BERT[2] and T5[3] became widely studied and applied[4][5][6]. In the era of large language models (LLMs), major advancements have led to the development of LLM-based embedding models, such as text-embedding-3-small/large (OpenAI), E5-Mistral-7B[28], SFR-Embedding-Mistral[29], SFR-Embedding-2R[30], GRITLM[31], LLM2Vec[12], RepLLaMA[10], BGE-en-icl[32], NV-Embed[19], gte-Qwen2-7B-Instruct[33], Qwen3-Embedding[34], etc. These models benefit from optimized LLM architectures—such as RoPE positional encoding[35], RMSNorm[36], and GeGLU activation[37]—combined with their strong semantic contextualization capabilities acquired through large-scale

pretraining. As a result, LLM-based embeddings achieve superior performance in retrieval and related tasks.

2.2 Embedding Model Training

The mainstream approaches currently involve contrastive learning pretraining on unsupervised/weakly supervised corpora and supervised contrastive learning training on high-quality labeled positive and negative samples. In unsupervised learning, early work like SimCSE[7] proposed feeding continuous inputs of both original and noise-augmented texts while employing contrastive learning to enhance the model’s discriminative representation capability. For weakly supervised learning, gte[33] utilized large-scale structured data (web search data, title-article pairs, etc.) for pretraining, followed by fine-tuning on high-quality open-source retrieval training data, achieving performance comparable to OpenAI embeddings with significantly fewer parameters. Conan-Embedding[24] and v2 similarly adopted the weakly supervised pretraining & supervised fine-tuning approach but incorporated techniques like cross-GPU batch loss balancing, dynamic hard negative mining, and soft masking (v2) to optimize the model. Seed1.6-Embedding[38] employed a phased training strategy combining text and multi-modal pretraining followed by business-scenario-specific fine-tuning, achieving superior representation quality.

Substantial research has also been conducted on modeling different tasks. Piccolo2[39] introduced multi-task hybrid loss functions for diverse downstream tasks, an approach we also incorporate. SFR-Embedding[30] utilized multi-task learning techniques to regularize embeddings, significantly enhancing domain data discrimination. Xiaobu-embedding unified the treatment of major CMTEB problem categories from the perspective of circle loss[40], fully leveraging multiple positive examples in original datasets while carefully balancing different loss weights.

2.3 Data Synthesis

Data quantity and quality are the most critical factors in model optimization, data synthesis methods have become a critical research direction due to the high cost of manual annotation. Doc2Query[41] and Query2Doc[42] employ question-answering models to generate pseudo-queries and pseudo-documents respectively, enhancing data for improved RAG performance. Promptagator[43] addresses few-shot retrieval scenarios by generating queries of diverse intents using few-shot demonstrations and annotations, effectively improving retrieval capabilities across varying intents or distributions. GPL[44] utilizes existing T5 encoder-decoder models to generate queries, retrieves similar passages as hard negatives using existing retrieval models, and employs cross-encoders to score each (query, passage) pair. Unnatural Instructions[45] leverages prompt and in-context learning (ICL) techniques to generate synthetic examples through controlled instructions, inputs, and constraints, producing 64k diverse data entries from several seed examples with promising experimental results. Qwen3-Embedding[34] designs a diversified prompting strategy by assigning document-specific roles to simulate potential users querying that document, enabling LLMs to generate stylistically authentic queries that enhance diversity and realism.

2.4 Hard Negative Mining Techniques

Hard negatives serve as essential components in contrastive learning for retrieval model training. Early work like ANCE[46] proposed an asynchronous ANN indexing mechanism that periodically updates hard negatives using checkpoint states to maintain optimally challenging samples. Conan-Embedding[24] and its v2 version implemented a dynamic hard negative sampling strategy by excluding and refreshing samples when their scores fall below a threshold. NV-Retriever[47] proposed positive-aware negative mining, introducing TopK-MarginPos and TopKPercPos filtering criteria to minimize false negatives. LGAI-Embedding[17] built upon NV-Retriever’s strategy with adaptive margin-based mining strategies, employing ANNA IR as a teacher retrieval model to identify high-quality hard negatives while using TopKPercPos filtering to eliminate false negatives.

3 Unified Multi-task Learning Framework

Embedding models support numerous downstream tasks including retrieval, reranking, STS, and classification. Given the diversity of these tasks and their associated data complexity, we explore a unified strategy to effectively handle them collectively while promoting optimization of the embedding model. Existing research on unified task processing includes circle loss[40], which approaches sentence pair similarity from a global perspective by categorizing tasks into class-level labels and pair-wise labels, Xiaobu-embedding demonstrated significant improvements by adopting this approach. Other models like Piccolo2[39], SFR-Embedding[30], NV-Embed[47], Conan-Embedding[24], and Conan-Embedding-v2 have incorporated multi-task learning using diverse training data with varying label processing methods, some employing task-specific losses (InfoNCE[48], Cosent[49], etc.).

Our design principle aims to accommodate more tasks and data types, enabling cross-domain and cross-task data to effectively enhance embedding capabilities. We propose a unified multi-task learning framework that categorizes training data into three task types: retrieval, NLI, and classification, with customized data and training solutions for each, allowing most natural text data to be converted into embedding training data through this framework. The following sections detail the framework’s components and implementation methods.

3.1 Model Architecture

Embedding models based on BERT or T5 [39][15][50][24] exhibit powerful contextual representation capabilities, primarily attributed to their bidirectional attention mechanisms. However, recent large language models predominantly adopt decoder-only architectures with unidirectional attention, significantly constraining tokens’ ability to capture contextual information. Several studies have addressed this limitation through architectural modifications or attention mechanism optimizations[12][31][47]. Our work builds upon the Qwen2.5-7B-Instruct architecture and checkpoint due to its exceptional Chinese language contextual capabilities. Consequently, we implemented the following modifications: (1) modifying the original causal attention to bi-directional attention

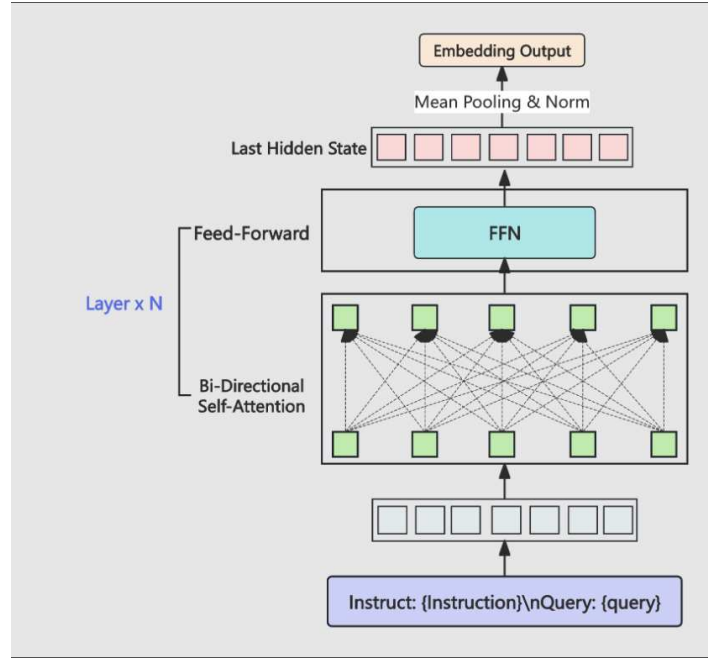


Figure 1: QZhou-Embedding Architecture

to enable comprehensive context capture, and (2) employing mean pooling with subsequent normalization to produce final embedding vectors. The model architecture is shown in Figure 1

3.2 Data Transformation

3.2.1 Retrieval-oriented Process

While open-source datasets such as MS MARCO[64] are readily accessible, they alone are insufficient for further advancing embedding model capabilities, thus we supplement with data from additional sources, such as news, academic paper and QA datasets. Given the heterogeneous nature of these datasets across domains and purposes, we design a retrieval-oriented data transformation methodology to convert diverse sources and formats into training data suitable for retrieval task. Below we outline selected categories of training data used for transformation and their processing procedures:

- **Title-Body/Abstract** "Title-Body/Abstract" type data primarily consists of title-body/article pairs typically sourced from online news, articles, documents, arXiv publications and Wikipedia. For these data types, the transformation process involves using the title as the query and the body/abstract as the positive sample. However, since the latter are documents, truncation is applied when they exceed the maximum training length.
- **Claim-Evidence** This data type typically presents a claim or statement followed by extracted evidence that either supports or refutes it, commonly used for multi-hop fact extraction and claim verification tasks. Datasets generally contain claims and corresponding evidence, with each evidence instance labeled as "Supports" or "Refutes". The transformation process involves: converting the claim portion

into a query sample, for evidence labeled as "Supports", the text is treated as a positive sample; for evidence labeled as "Refutes", it is converted into a negative sample.

- **Question-Answer** Question-answering data and conversational Q-A pairs primarily originate from chat platforms and forums. Within the current wave of LLM and reinforcement learning research, such data exhibits remarkable volume and diversity. Virtually single-turn Q-A datasets(one question paired with one answer) represents the most suitable format for retrieval training. For transformation, the "Question/Query/User" portion is converted into queries, while the "Answer/Response/Assistant" portion is processed as documents.

3.2.2 NLI-oriented Process

Natural Language Inference (NLI) represents a fundamental capability of NLP models, encompassing tasks such as semantic similarity, textual entailment, and sentiment analysis. This section describes the methodology for transforming and constructing training sets from NLI-style data, using textual semantic similarity (STS) and textual entailment tasks as illustrative examples. Our approach distinctively reformulates NLI tasks into text_pair-score formats compatible with Cosent loss[49] training strategy, where sample pairs are quantitatively scored based on their semantic relationships. The processing procedures for each are detailed below:

- **STS** Semantic Textual Similarity (STS) is characterized by its symmetric semantic matching to determine whether two sentences share equivalent meaning. STS datasets typically consist of sentence pairs with associated labels, which may be binary classifications (yes/no, true/false) or numerical scores (e.g., 1.2, 3.1, 4.8). For binary labels, "yes"/"true" are mapped to a numerical value of 1, while "no"/"false" are converted to 0. The data is then structured into (query, document, score) triplets. Due to the symmetric nature of STS, each single original data sample can generate two training triplets by interchanging the query and positive document roles.
- **Textual Entailment** Textual entailment further examines a model's capabilities in reasoning, typically featuring three-class labels: entailment, neutral, contradiction. Our processing method employs a three-tier scoring system: labels are assigned values of 2, 1, and 0 for entailment, neutral, and contradiction respectively. We construct (query, document, score) triplets accordingly, and similarly leverage symmetry to double the dataset size.

3.2.3 CLS-oriented Process

Classification tasks encompass text categorization and sentiment classification scenarios, it typically follows a (text, label) format, where texts within the same category exhibit semantic proximity while distinct boundaries separate different classes. NV-Embed[47] compared label-based and example-based data construction methods, with experimental results demonstrating the superiority of the latter. Adopting the example-based approach, we process classification data (text, label) by using the text as query,

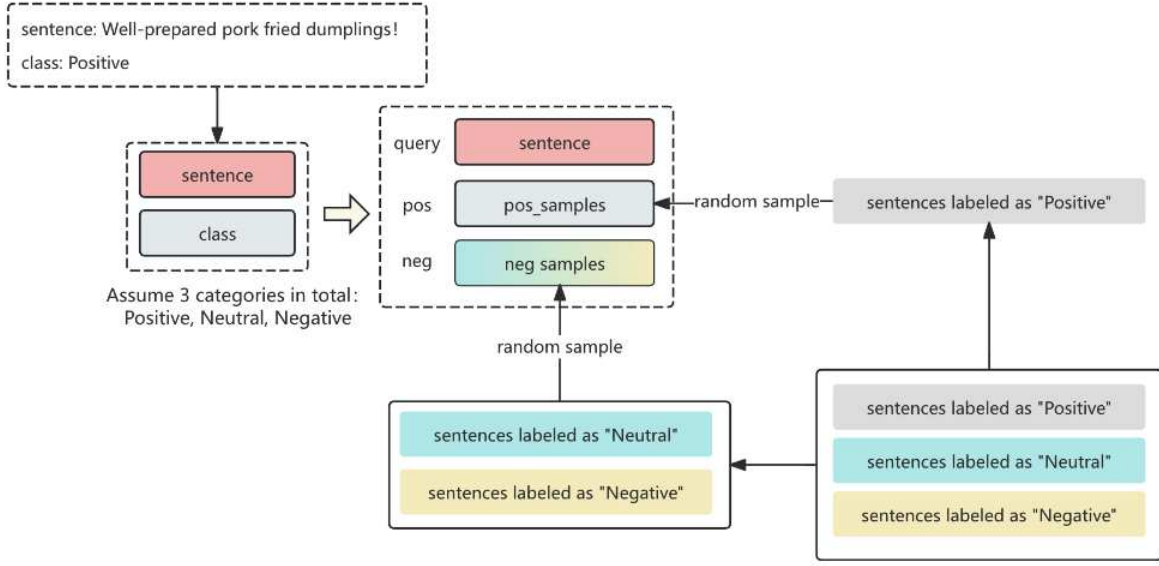


Figure 2: CLS-oriented data transformation

sampling other texts sharing the same label as positive examples, and selecting texts from different labels as negative examples. Figure 2 provides a detailed schematic illustration of this process.

3.3 Training Strategy

Each task category—retrieval, NLI, and classification—operates within a data construction process respectively, for which we have designed specialized training objectives to enhance model training efficiency. This section elaborates on the design of loss functions for retrieval, NLI, and classification tasks.

3.3.1 Retrieval

For the retrieval task, we adopt the widely used InfoNCE loss[48], but incorporate an improvement inspired by gte[33] by augmenting the original query-negative loss with an additional query-query loss term. Specifically, each query within a batch is treated as a negative sample for all other queries. The final loss formulation is explicitly described in Equation (1).

$$\mathcal{L}_{\text{Retrieval}} = -\frac{1}{n} \sum_i \log \frac{e^{\text{sim}(q_i, d_i^+)/\tau}}{e^{\text{sim}(q_i, d_i^+)/\tau} + \sum_j e^{\text{sim}(q_i, d_j^-)/\tau} + \sum_{j \neq i} e^{\text{sim}(q_i, q_j)/\tau}} \quad (1)$$

3.3.2 NLI

For NLI tasks, the transformed labels are numerically comparable and exhibit ordinal relationships. We employ Cosent loss[49] to optimize such data, which is designed based on the principles of Circle loss[40]. As a ranking-sensitive loss function, Cosent loss requires only ordinal label information for optimization while demonstrating faster convergence. Its mathematical formulation is presented in Equation (2).

$$\mathcal{L}_{\text{NLI}} = \log(1 + \sum_{\text{sim}(i,j) > \text{sim}(k,l)} \exp(\frac{\text{sim}(x_k, x_l) - \text{sim}(x_i, x_j)}{\tau})) \quad (2)$$

3.3.3 CLS

The classification loss also adopts the InfoNCE objective. However, since CLS data is processed in an example-based manner, directly applying in-batch negative sampling on classification datasets with limited categories may lead to false negatives from items of different classes. Numerous studies have proposed diverse approaches to address this issue[51][52][47]. We propose a masking mechanism that appends class labels to each positive and negative sample during preprocessing (recorded as separate variables rather than modifying raw text). During in-batch negative sampling, for each negative sample from other data instances, we check whether its label matches the current query’s class. If matched, the negative loss contribution is masked to zero to prevent erroneous penalization; otherwise, it is normally computed. The core loss remains InfoNCE, with the CLS loss formulation shown in Equation (3). Where C_{t_i} denotes the class label of sample t_i , and n represents the number of negative samples per data instance.

$$L_{\text{CLS}} = -\frac{1}{n} \sum_i \log \frac{e^{\text{sim}(t_i, t_i^+)/\tau}}{Z_i} \quad (3)$$

$$\begin{aligned} \text{where } Z_i &= e^{\text{sim}(t_i, t_i^+)/\tau} + \sum_n \text{MASK}(t_i, t_{i,n}^-) \cdot e^{\text{sim}(t_i, t_{i,n}^-)/\tau} + \\ &\sum_{j \neq i} \text{MASK}(t_i, t_j) \cdot e^{\text{sim}(t_i, t_j)/\tau} + \\ &\sum_{j \neq i} \sum_n \text{MASK}(t_i, t_{j,n}^-) \cdot e^{\text{sim}(t_i, t_{j,n}^-)/\tau} \end{aligned}$$

$$\text{and } C_{t_i} = C_{t_i^+}$$

$$\text{and } \text{MASK}(t_i, t_j) = \begin{cases} 0 & \text{if } C_{t_i} = C_{t_j}, \\ 1 & \text{otherwise} \end{cases}$$

4 Data Synthesis

The production of higher-quality data through data production has gained critical importance in embedding training. Manual annotation incurs higher costs and lower production efficiency, thus developing effective automated data synthesis methods has emerged as a key research focus. Recent advancements in large language models (LLMs) have significantly improved their linguistic capabilities, enabling accurate interpretation of human instructions and generation of high-quality outputs. Multiple existing methods have effectively leveraged LLMs to generate high-quality data[28][34], we similarly

leverages LLM capabilities for data production across three dimensions: structural diversity, semantic diversity, and difficulty, with dedicated synthesis strategies for each. For structural diversity, we propose Paraphrasing techniques; for semantic diversity, we introduce Augmentation methods; and to increase training difficulty and improve semantic discriminability, we employ LLMs to generate more challenging hard negative examples. The following sections detail these methodologies. The constraint components for all data synthesis techniques are specified in Table 5 of Appendix A.1.

4.1 Structural Diversity Enhancement

Linguistic structures of text encompass lexical, syntactic, and grammatical features, which represent relatively surface-level characteristics reflecting word arrangements, combinations, tenses, voices, and other formal attributes. Embedding models must accurately capture underlying semantics despite variations in surface form, ensuring robustness to external structural changes. For example, the following two sentences, despite structural differences, should be recognized as semantically equivalent:

- The cat chased the mouse.
- The mouse was chased by the cat.

To effectively train an embedding model that remains invariant to structural variations while accurately capturing semantic information, we propose a Paraphrasing strategy. For each training sample containing a query and a positive document, we apply LLM-based paraphrasing to both contents, generating augmented instances that preserve semantic equivalence while introducing structural divergence. The prompt constraints and workflow are illustrated in Figure 3.

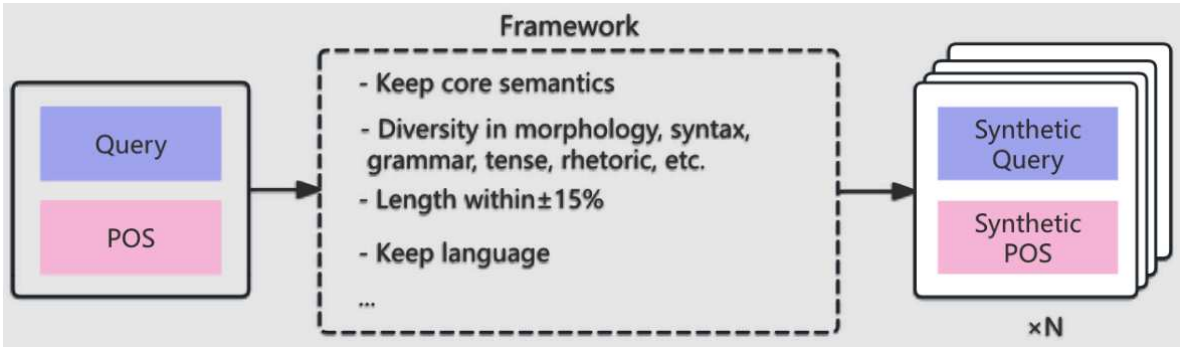


Figure 3: LLM-based Paraphrasing Workflow

4.2 Semantic Diversity Enhancement

Merely augmenting data through superficial structural modifications yields negligible improvements in model capabilities, as generalization relies not only on structural disentanglement but also on diverse topics and content to ensure uniform vector representations in the spatial domain. Therefore, beyond paraphrasing, we propose an augmentation method using LLM to diversify semantics. The core concept is: given a

complete (query, positive) pair, the model must comprehend the domain and perspective discussed and learn to expand into different topics, aspects, and viewpoints while remaining contextually anchored. This process is governed via prompt constraints. The Augmentation framework is illustrated in Figure 4.

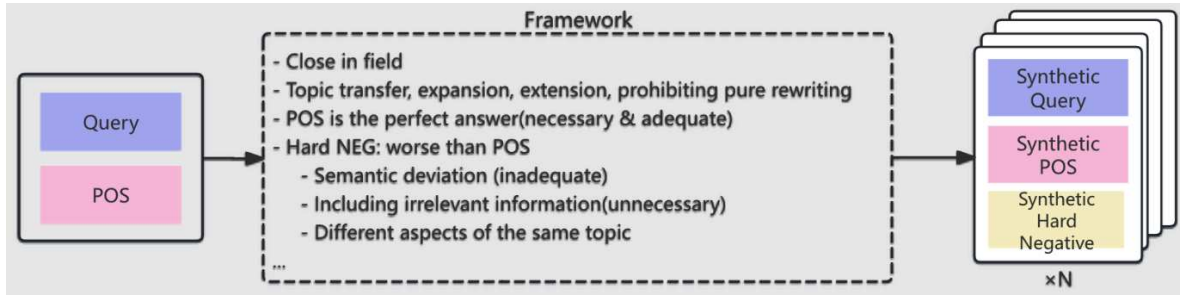


Figure 4: Semantic Augmentation Workflow

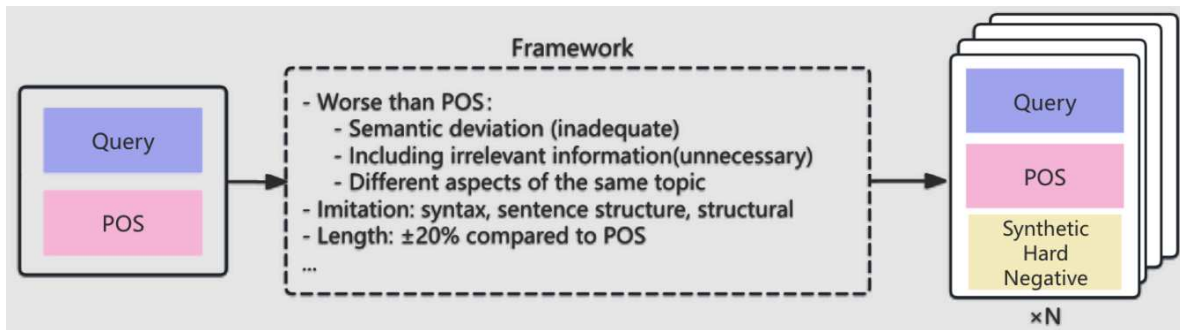


Figure 5: Hard Negative Synthesis Workflow

4.3 More challenging embeddings

Hard negative examples are crucial for enhancing the performance of text embedding models, often requiring substantial effort to acquire. Leveraging the linguistic capabilities of large language models, we design an automated hard negative synthesis method tailored for retrieval datasets. Our domain-specific experiments demonstrate that large language models can generate examples that are indistinguishable, the framework is illustrated in Figure 5.

During Data paraphrasing and Augmentation, we implement task-specific strategies: for retrieval tasks, we rewrite/expand (query, positive) pairs and add them to the original dataset; for NLI tasks, we rewrite individual sentences by randomly duplicating existing entries containing the original sentences and replacing them with rewritten versions to achieve data expansion—without applying augmentation to prevent ambiguity; for classification tasks, we rewrite sentences while retaining their original labels, example-based processing was applied using the rewritten results, again without employing augmentation. We provide several data synthesis examples in Appendix A.3 for reference.

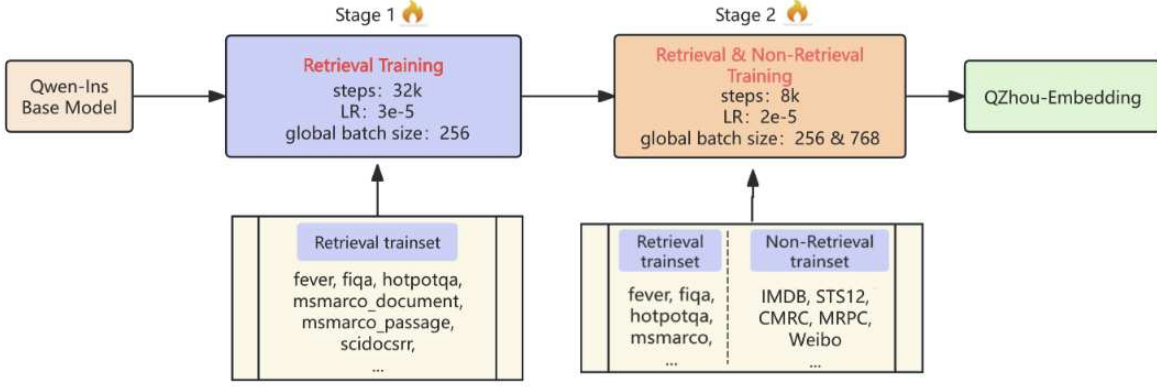


Figure 6: Training pipeline

5 Training Optimization

5.1 Data Grouping Strategy

Prior works like Linq-Embedding[52] and SFR-Embedding-Mistral[30] adopted task-homogeneous batching, partitioning data by task rather than mixing them, and sampling tasks based on weighted randomness during training. Building on this, we propose a refined Data Grouping Strategy, extending the granularity from task-level to dataset-level partitioning. We posit that dataset-level grouping captures more domain-specific clustering patterns—samples within the same dataset often exhibit inherent domain similarities, while such consistency may not hold across datasets.

Our approach partitions training data into subsets by name. During training, only samples from a single dataset are sampled per batch, with file pointers recorded to enable sequential reading in subsequent iterations. For sampling weights, we adopt the data sampling strategy from gte[33] and mgte[50], scaling weights by dataset size followed by normalization. For dataset i with size l_i , its sampling weight is computed as Equation (4)

$$p_i = \frac{l_i^\alpha}{\sum_{j=1}^m l_j^\alpha} \quad (4)$$

5.2 Two-Stage Training

Inspired by NV-Embed’s[47] two-stage contrastive learning instruction tuning technique, we adopt a similar training approach: the first stage exclusively uses retrieval-oriented training data, while the second stage integrates both retrieval and non-retrieval tasks, the overall training framework is illustrated in the figure 6. Two key distinctions are incorporated: first, we integrate the previously described Data Grouping Strategy; second, we implement global control over the sampling ratio of retrieval training datasets, since our findings indicate that naively incorporating additional data significantly degrades retrieval performance.

For global control of sampling ratio, a hyperparameter η is introduced into the sampling

function to control the proportion of retrieval training, ensuring that throughout the second training stage, the computational contribution of retrieval data accounts for η , while non-retrieval data constitutes $1 - \eta$. The following set of equations formalizes the computational process from partitioned datasets to sampling ratio determination. Let the training data $D = [d_1, d_2, \dots, d_N]$, where each d_i represents a distinct dataset (e.g., MSMARCO_passage, SQUAD), with corresponding sizes $L = [l_1, l_2, \dots, l_N]$. Following the aforementioned strategy, we first apply an exponential scaling factor α , a mask factor M is then applied to filter retrieval and non-retrieval training sets for summation. The equations are as follows:

$$S_{ret} = \sum_i M_i \cdot l_i^\alpha$$

$$S_{non_ret} = \sum_i (1 - M_i) \cdot l_i^\alpha$$

$$where\ M_i = \begin{cases} 0 & \text{if } d_i \in \text{RET}, \\ 1 & \text{else} \end{cases}$$

where RET denotes the set of retrieval training datasets. The retrieval ratio is then scaled using η to derive the final normalized sampling ratios for the training sets:

$$L_{samp} = [l_1^{samp}, l_2^{samp}, \dots, l_N^{samp}]$$

$$where\ l_i^{samp} = \begin{cases} \frac{\eta_{RET} \cdot l_i^\alpha}{S_{ret}} & \text{if } d_i \in \text{RET}, \\ \frac{(1-\eta_{RET}) \cdot l_i^\alpha}{S_{non_ret}} & \text{else} \end{cases}$$

6 Experiments

6.1 Training Dataset

Primary data sources include bge-en-icl, bge-m3-data, and bge-multilingual-gemma2-data³. The E5 dataset (approximately 1.5M samples)⁴, utilized in E5-Mistral-7B[28], Echo Embedding[11], and LLM2Vec[12], is also incorporated. The aforementioned datasets include commonly used retrieval training corpora such as MS MARCO (both passage and document versions)[64], Natural Questions (NQ)[65], ELI5[66], HotpotQA[67], MIRACL[68], SQuAD[69], FEVER[70], Quora Question Pairs(QQP), and DuReader[71], etc. Previous researchers have already systematically collected and organized these datasets, making them readily usable, we solely utilized the proposed method to update harder negative samples. Stella’s[53] retrieval_data_llm⁵ provides high-quality (query, positive, negative) triplets, while zpoint leverages datasets such as Huatuo medical QA⁶, all above data has been incorporated. Additional data from huggingface’s sentence-transformers⁷ repository includes reddit, hover[72], mr-tydi[73], law-gpt, and s2orc[74].

³<https://github.com/FlagOpen/FlagEmbedding/tree/master/dataset>

⁴https://drive.google.com/file/d/1YqgaJIzmBIH37XBxpRPCVzV_CLh6aOI4/view

⁵https://huggingface.co/datasets/infracol/retrieval_data_llm

⁶https://huggingface.co/iampanda/zpoint_large_embedding_zh

⁷<https://huggingface.co/sentence-transformers>

Other sources encompass web_questions, BioASQ[54], cmrc[55], CSL⁸, nli_for_simcse (used in SimCSE[7] and GTE[33]), MLDR⁹, GLUE Benchmark[56], Yelp Reviews[57] and Weibo Sentiment¹⁰ training sets.

We further integrate MTEB evaluation-related datasets like Imdb-Classification[58], MassiveIntent-Classification[59], MassiveScenario-Classification[59], STS12[60], LCQMC[61], PAWSX[62], and STSB[63], we utilized the training split from these datasets with contamination exclusion applied to remove samples highly similar to test sets.

For data requiring format conversion, we apply the methodologies described in Section 3.2. Datasets with limited samples (e.g., subsets of bge and e5 series, Imdb-Classification, STS12, LCQMC) are augmented via Paraphrasing and Augmentation (typically applied to datasets with fewer than 60k samples), we ultimately obtained approximately 5M high-quality training samples through API interfaces. We deduplicate all training sets and filter out samples with low query-pos scores using GTE-Qwen2-7B-Instruct¹¹. For retrieval data lacking hard negatives, we employ synthetic hard negative generation. Due to API cost constraints, only 30% of hard negatives are synthetically generated; the remainder are produced using stella-large-zh-v3-1792d[53], with top-10 to top-30 ranked results selected as hard negatives. The final training dataset contains 11M quadruples (query, pos, neg, instruction) in total.

6.2 Trainset Instructions

For most training data containing instruction formats, we retain their original contents. For the MTEB training set, we adopt instructions corresponding to its evaluation (consistent with Qwen3-Embedding runtime). For external data lacking instructions (e.g., Huatuo, Reddit, Law-GPT, GLUE), we design task-specific and domain-adaptive instructions. Partial instruction templates are provided in Appendix A.2.

6.3 Training Details

As previously mentioned, we adopt a two-stage training approach. For the first-stage retrieval training, we train on all retrieval datasets, with a warm-up step of 300 and a learning rate of $3e-5$, the total step of training is 32k. In the second stage, we use all training data, set the learning rate to $2e-5$, and train for 8k steps, keeping all other configurations the same as in the first stage. We employ a batch size of 256 for all data using the InfoNCE loss (i.e., retrieval and classification), considering data using the cosent loss (i.e., NLI), due to lower memory consumption from the absence of forward computation for negative samples, the batch size is set to 768. Across all stages, we employ bfloat16 precision, with 4 hard negative samples and a cosine temperature of 0.02, using Adam optimizer with a weight decay of 0.01. The Data Grouping Strategy remains unchanged between the two stages, except that the second stage incorporates all data with a global retrieval ratio η_{RET} of 0.72. Unlike existing works that commonly

⁸<https://github.com/ydli-ai/CSL?tab=readme-ov-file>

⁹<https://huggingface.co/datasets/Shitao/MLDR>

¹⁰<https://github.com/SophonPlus/ChineseNlpCorpus?tab=readme-ov-file>

¹¹<https://huggingface.co/Alibaba-NLP/gte-Qwen2-7B-instruct>

use LoRA fine-tuning, we employ full-parameter fine-tuning at all stages to ensure maximum performance improvement. The query and passage lengths are set to 256 and 1536 respectively. However, in practice, the model can handle sequences up to 8k in length due to the strong length extrapolation capability of the RoPE[35] positional encoding used in most LLMs. The hyperparameter configurations for all training stages are provided in the table 1.

Table 1: Training Hyperparameter Specifications

Item	Stage1	Stage2
Warm-up	300	
Steps	3e-5	2e-5
LR	32k	8k
Batch Size InfoNCE	256	
Batch Size Cosent	-	768
Precision	bfloat16	
Temperature	0.02	
Optimizer	Adam	
Query Length	256	
Passage Length	1536	

6.4 Compared Methods

We selected the top-10 ranked models(August 27, 2025) on the MTEB/CMTEB leaderboards prior to the release of QZhou-Embedding as baselines. For MTEB, the comparative models include LGAI-Embedding-Preview[17], the Seed series (v1.5[75] , v1.6[38]), Qwen series (8B, 4B)[34], retrieve_zh_v1, xiaobu-embedding-v2, gemini-embedding-001[76], jasper_en_vision_language_v1[14], Linq-Embed-Mistral[52], SFR-Embedding-Mistral[30], and NV-Embed-v2[47]. For CMTEB, the baseline models comprise the Seed series (as above), Qwen series (as above), Conan series (v1[24], v2[13]), zpoint_large_embedding_zh, and piccolo-large-zh-v2[39].

6.5 Main Results

This section presents the evaluation results of Qzhou-embedding on MTEB/CMTEB benchmarks, alongside comparative scores from the top 10 ranked models. As detailed in Table 2, Table 3, Qzhou-embedding achieves state-of-the-art performance across both task-level and task-type average metrics, demonstrating the effectiveness of our approach. Furthermore, under MTEB’s official ranking protocol, Qzhou-embedding secured the top position on both leaderboards. (**Note:** Highlighted maximum values in certain columns may reflect the best performance among the listed models rather than the overall leaderboard maximum, as exemplified by the MTEB/classification benchmark where the top score does not appear in the top 10 models.)

Table 2: Performance on MTEB(eng, v2)

Model	Class.	Clust.	Pair Class.	Rerank.	STS	Retr.	Summ.	Mean(Task)	Mean(TaskType)
LGAI-Embedding-Preview	89.97	59.25	88.67	49.13	66.18	86.69	38.93	74.12	68.4
Seed1.5-Embedding	89.88	60.83	87.39	50.67	67.45	87.23	36.44	74.76	68.56
Qwen3-Embedding-8B	90.43	58.57	87.52	51.56	69.44	88.58	34.83	75.22	68.71
Qwen3-Embedding-4B	89.84	57.51	87.01	50.76	68.46	88.72	34.39	74.6	68.1
Seed1.6-embedding	92.42	59.22	85.07	50.28	64.9	86.87	37.1	74.07	67.98
gemini-embedding-001	90.05	59.39	87.7	48.59	64.35	85.29	38.28	73.3	67.67
jasper_en_vision_language_v1	90.27	60.52	88.14	50	56.05	84.37	37.19	71.41	66.65
Linq-Embed-Mistral	83	54.07	88.44	49.44	60.14	84.69	37.26	69.8	65.29
SFR-Embedding-Mistral	80.47	54.93	88.59	50.15	59.33	84.77	36.32	69.31	64.94
NV-Embed-v2	87.19	47.66	88.69	49.61	62.84	83.82	35.21	69.81	65
QZhou-Embedding(Ours)	88.97	61.65	92.43	51.77	67.12	91.65	33.05	75.97	69.52

Table 3: Performance on CMTEB(cmn, v1)

Model	Class.	Clust.	Pair Class.	Rerank.	STS	Retr.	Mean(Task)	Mean(TaskType)
Seed1.6-embedding	77.98	73.11	88.71	71.65	79.69	68.94	75.63	76.68
Seed1.5-Embedding	79.37	71.11	89.57	70.14	79.33	66.56	74.87	76.01
retrieve.zh.v1	76.88	66.5	85.98	72.86	76.97	63.92	72.71	73.85
Conan-embedding-v2	76.47	68.84	92.44	74.41	78.31	65.48	74.24	75.99
xiaobu-embedding-v2	76.53	65.17	85.94	72.58	76.49	64.18	72.36	73.48
Qwen3-Embedding-8B	76.97	80.08	84.23	66.99	78.21	63.53	73.84	75
Conan-embedding-v1	76.77	66.33	85.68	72.76	76.67	63.67	72.5	73.65
zpoint_large_embedding_zh	76.4	62.23	85.75	72.33	76.36	63.86	71.81	72.82
piccolo-large-zh-v2	76.42	62.16	85.22	70	74.36	63.46	70.86	71.94
Qwen3-Embedding-4B	75.46	77.89	83.34	66.05	77.03	61.26	72.27	73.51
QZhou-Embedding(Ours)	79.99	70.91	95.07	74.85	78.80	71.89	76.99	78.58

7 Conclusion

In this technical report, we present QZhou-Embedding, a general-purpose contextual text embedding model with exceptional text representation capabilities. We designed a unified multi-task framework comprising specialized data transformation and training strategies, effectively enhanced the diversity of training data. To further improve the quality of training data and the model’s generalization capabilities, we developed a data synthesis pipeline leveraging LLM API, incorporating techniques such as Paraphrasing, Augmentation, and Hard negative example generation. We employ a two-stage training strategy comprising initial retrieval-focused training followed by full-task fine-tuning, enabling the embedding model to extend its capabilities based on robust retrieval performance. The model achieves state-of-the-art results on the MTEB and CMTEB benchmarks, ranking first on both leaderboards. Our findings establish that data quality and diversity are pivotal for improving embedding model capabilities. In the future, we will focus on developing multimodal and multilingual embedding models, as well as exploring effective applications of embedding models in agent systems, aiming to integrate cutting-edge technologies to optimize this classical module.

References

- [1] Robertson, Stephen E., and Steve Walker. "Some simple effective approximations to the 2-poisson model for probabilistic weighted retrieval." In SIGIR'94: Proceedings

- of the Seventeenth Annual International ACM-SIGIR Conference on Research and Development in Information Retrieval, organised by Dublin City University, pp. 232-241. London: Springer London, 1994.
- [2] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. Bert: Pre-training of deep bidirectional transformers for language understanding. arXiv preprint arXiv:1810.04805, 2018.
 - [3] Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J Liu. Exploring the limits of transfer learning with a unified text-to-text transformer. *Journal of machine learning research*, 21(140):1–67, 2020.
 - [4] Liang Wang, Nan Yang, Xiaolong Huang, Binxing Jiao, Linjun Yang, Daxin Jiang, Rangan Majumder, and Furu Wei. Text embeddings by weakly-supervised contrastive pre-training. arXiv preprint arXiv:2212.03533, 2022.
 - [5] Gautier Izacard, Mathilde Caron, Lucas Hosseini, Sebastian Riedel, Piotr Bojanowski, Armand Joulin, and Edouard Grave. Unsupervised dense information retrieval with contrastive learning. arXiv preprint arXiv:2112.09118, 2021.
 - [6] Nils Reimers and Iryna Gurevych. Sentence-bert: Sentence embeddings using siamese bert-networks. arXiv preprint arXiv:1908.10084, 2019.
 - [7] Tianyu Gao, Xingcheng Yao, and Danqi Chen. 2021. SimCSE: Simple contrastive learning of sentence embeddings. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 6894–6910, Online and Punta Cana, Dominican Republic. Association for Computational Linguistics.
 - [8] Jianmo Ni, Chen Qu, Jing Lu, Zhuyun Dai, Gustavo Hernández Ábrego, Ji Ma, Vincent Y Zhao, Yi Luan, Keith B Hall, Ming-Wei Chang, et al. Large dual encoders are generalizable retrievers. arXiv preprint arXiv:2112.07899, 2021.
 - [9] Brown, Tom, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D. Kaplan, Prafulla Dhariwal, Arvind Neelakantan et al. "Language models are few-shot learners." *Advances in neural information processing systems* 33 (2020): 1877-1901.
 - [10] Ma, Xueguang, Liang Wang, Nan Yang, Furu Wei, and Jimmy Lin. "Fine-tuning llama for multi-stage text retrieval." In *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval*, pp. 2421-2425. 2024.
 - [11] Springer, Jacob Mitchell, Suhas Kotha, Daniel Fried, Graham Neubig, and Aditi Raghunathan. "Repetition improves language model embeddings." arXiv preprint arXiv:2402.15449 (2024).
 - [12] Behnam Ghader, Parishad, Vaibhav Adlakha, Marius Mosbach, Dzmitry Bahdanau, Nicolas Chapados, and Siva Reddy. "Llm2vec: Large language models are secretly powerful text encoders." arXiv preprint arXiv:2404.05961 (2024).
 - [13] <https://cloud.tencent.com/developer/news/2461911>

- [14] Zhang, Dun, Jiacheng Li, Ziyang Zeng, and Fulong Wang. "Jasper and stella: distillation of sota embedding models." arXiv preprint arXiv:2412.19048 (2024).
- [15] Chen, Jianlv, Shitao Xiao, Peitian Zhang, Kun Luo, Defu Lian, and Zheng Liu. "Bge m3-embedding: Multi-lingual, multi-functionality, multi-granularity text embeddings through self-knowledge distillation." arXiv preprint arXiv:2402.03216 (2024).
- [16] Ji, Yifan, Zhipeng Xu, Zhenghao Liu, Yukun Yan, Shi Yu, Yishan Li, Zhiyuan Liu, Yu Gu, Ge Yu, and Maosong Sun. "Learning more effective representations for dense retrieval through deliberate thinking before search." arXiv preprint arXiv:2502.12974 (2025).
- [17] Choi J, Kim H, Jang H, et al. LG-ANNA-Embedding technical report[J]. arXiv preprint arXiv:2506.07438, 2025.
- [18] Xiong, Lee, Chenyan Xiong, Ye Li, Kwok-Fung Tang, Jialin Liu, Paul Bennett, Junaid Ahmed, and Arnold Overwijk. "Approximate nearest neighbor negative contrastive learning for dense text retrieval." arXiv preprint arXiv:2007.00808 (2020).
- [19] Lee, Chankyu, Rajarshi Roy, Mengyao Xu, Jonathan Raiman, Mohammad Shoeybi, Bryan Catanzaro, and Wei Ping. "Nv-embed: Improved techniques for training llms as generalist embedding models." arXiv preprint arXiv:2405.17428 (2024).
- [20] Moreira, Gabriel de Souza P., Radek Osmulski, Mengyao Xu, Ronay Ak, Benedikt Schifferer, and Even Oldridge. "NV-Retriever: Improving text embedding models with effective hard-negative mining." arXiv preprint arXiv:2407.15831 (2024).
- [21] Team, Qwen. "Qwen2 technical report." arXiv preprint arXiv:2407.10671 (2024).
- [22] Xiao, Shitao, Zheng Liu, Peitian Zhang, Niklas Muennighoff, Defu Lian, and Jian-Yun Nie. "C-pack: Packed resources for general chinese embeddings." In Proceedings of the 47th international ACM SIGIR conference on research and development in information retrieval, pp. 641-649. 2024. Team, Qwen.
- [23] Muennighoff, Niklas, Nouamane Tazi, Loïc Magne, and Nils Reimers. "Mteb: Massive text embedding benchmark." arXiv preprint arXiv:2210.07316 (2022).
- [24] Li, Shiyu, Yang Tang, Shizhe Chen, and Xi Chen. "Conan-embedding: General text embedding with more and better negative samples." arXiv preprint arXiv:2408.15710 (2024).
- [25] Aizawa, Akiko. "An information-theoretic perspective of tf-idf measures." Information Processing & Management 39, no. 1 (2003): 45-65.
- [26] Robertson, Stephen E., and Steve Walker. "Some simple effective approximations to the 2-poisson model for probabilistic weighted retrieval." In SIGIR'94: Proceedings of the Seventeenth Annual International ACM-SIGIR Conference on Research and Development in Information Retrieval, organised by Dublin City University, pp. 232-241. London: Springer London, 1994.

- [27] Deerwester, Scott, Susan T. Dumais, George W. Furnas, Thomas K. Landauer, and Richard Harshman. "Indexing by latent semantic analysis." *Journal of the American society for information science* 41, no. 6 (1990): 391-407.
- [28] Liang Wang, Nan Yang, Xiaolong Huang, Linjun Yang, Rangan Majumder, and Furu Wei. Improving text embeddings with large language models. *arXiv preprint arXiv:2401.00368*, 2023b.
- [29] Meng, Rui, Ye Liu, Shafiq Rayhan Joty, Caiming Xiong, Yingbo Zhou, and Semih Yavuz. "Sfembedding-mistral: enhance text retrieval with transfer learning." *Salesforce AI Research Blog* 3 (2024): 6.
- [30] Meng R, Liu Y, Joty S R, et al. Sfr-embedding-2: Advanced text embedding with multi-stage training, 2024[J].
- [31] Muennighoff, Niklas, S. U. Hongjin, Liang Wang, Nan Yang, Furu Wei, Tao Yu, Amanpreet Singh, and Douwe Kiela. "Generative representational instruction tuning." In *The Thirteenth International Conference on Learning Representations*. 2024.
- [32] Chaofan Li, MingHao Qin, Shitao Xiao, Jianlyu Chen, Kun Luo, Yingxia Shao, Defu Lian, and Zheng Liu. Making text embedders few-shot learners. *arXiv preprint arXiv:2409.15700*, 2024.
- [33] Zehan Li, Xin Zhang, Yanzhao Zhang, Dingkun Long, Pengjun Xie, and Meishan Zhang. Towards general text embeddings with multi-stage contrastive learning, 2023. URL <https://arxiv.org/abs/2308.03281>.
- [34] Zhang, Yanzhao, Mingxin Li, Dingkun Long, Xin Zhang, Huan Lin, Baosong Yang, Pengjun Xie et al. "Qwen3 Embedding: Advancing Text Embedding and Reranking Through Foundation Models." *arXiv preprint arXiv:2506.05176* (2025).
- [35] Su, Jianlin, Murtadha Ahmed, Yu Lu, Shengfeng Pan, Wen Bo, and Yunfeng Liu. "Roformer: Enhanced transformer with rotary position embedding." *Neurocomputing* 568 (2024): 127063.
- [36] Zhang, Biao, and Rico Sennrich. "Root mean square layer normalization." *Advances in neural information processing systems* 32 (2019).
- [37] Shazeer, Noam. "Glu variants improve transformer." *arXiv preprint arXiv:2002.05202* (2020).
- [38] <https://seed1-6-embedding.github.io/>
- [39] Huang, Junqin, Zhongjie Hu, Zihao Jing, Mengya Gao, and Yichao Wu. "Piccolo2: General text embedding with multi-task hybrid loss training." *arXiv preprint arXiv:2405.06932* (2024).
- [40] Sun, Yifan, Changmao Cheng, Yuhao Zhang, Chi Zhang, Liang Zheng, Zhongdao Wang, and Yichen Wei. "Circle loss: A unified perspective of pair similarity optimization." In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 6398-6407. 2020.

- [41] Rodrigo Nogueira, Wei Yang, Jimmy Lin, and Kyunghyun Cho. 2019. Document expansion by query prediction. ArXiv preprint, abs/1904.08375.
- [42] Liang Wang, Nan Yang, and Furu Wei. 2023. Query2doc: Query expansion with large language models. In Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing, pages 9414–9423, Singapore. Association for Computational Linguistics.
- [43] Zhuyun Dai, Vincent Y Zhao, Ji Ma, Yi Luan, Jianmo Ni, Jing Lu, Anton Bakalov, Kelvin Guu, Keith Hall, and Ming-Wei Chang. 2022. Promptagator: Fewshot dense retrieval from 8 examples. In The Eleventh International Conference on Learning Representations.
- [44] Kexin Wang, Nandan Thakur, Nils Reimers, and Iryna Gurevych. 2022a. GPL: Generative pseudo labeling for unsupervised domain adaptation of dense retrieval. In Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, pages 2345–2360, Seattle, United States. Association for Computational Linguistics.
- [45] Honovich, Or, Thomas Scialom, Omer Levy, and Timo Schick. "Unnatural instructions: Tuning language models with (almost) no human labor." arXiv preprint arXiv:2212.09689 (2022).
- [46] Xiong, Lee, Chenyan Xiong, Ye Li, Kwok-Fung Tang, Jialin Liu, Paul Bennett, Junaid Ahmed, and Arnold Overwijk. "Approximate nearest neighbor negative contrastive learning for dense text retrieval." arXiv preprint arXiv:2007.00808 (2020).
- [47] Moreira, Gabriel de Souza P., Radek Osmulski, Mengyao Xu, Ronay Ak, Benedikt Schifferer, and Even Oldridge. "NV-Retriever: Improving text embedding models with effective hard-negative mining." arXiv preprint arXiv:2407.15831 (2024).
- [48] Aaron van den Oord, Yazhe Li, and Oriol Vinyals. Representation learning with contrastive predictive coding. arXiv preprint arXiv:1807.03748, 2018.
- [49] <https://www.kexue.fm/archives/8847>
- [50] Xin Zhang, Yanzhao Zhang, Dingkun Long, Wen Xie, Ziqi Dai, Jialong Tang, Huan Lin, Baosong Yang, Pengjun Xie, Fei Huang, Meishan Zhang, Wenjie Li, and Min Zhang. mgte: Generalized long-context text representation and reranking models for multilingual text retrieval, 2024.
- [51] Lee, Jinhyuk, Zhuyun Dai, Xiaoqi Ren, Blair Chen, Daniel Cer, Jeremy R. Cole, Kai Hui et al. "Gecko: Versatile text embeddings distilled from large language models, 2024." URL <https://arxiv.org/abs/2403.20327>.
- [52] Junseong Kim, Seolhwa Lee, Jihoon Kwon, Sangmo Gu, Yejin Kim, Minkyung Cho, Jy yong Sohn, and Chanyeol Choi. Linq-embed-mistral: Elevating text retrieval with improved gpt data through task-specific control and quality refinement. linq ai research blog, 2024.
- [53] <https://huggingface.co/dunzhang/stella-large-zh-v3-1792d>

- [54] Tsatsaronis G, Balikas G, Malakasiotis P, et al. An overview of the BIOASQ large-scale biomedical semantic indexing and question answering competition[J]. BMC bioinformatics, 2015, 16(1): 138.
- [55] Cui Y, Liu T, Che W, et al. A span-extraction dataset for Chinese machine reading comprehension[J]. arXiv preprint arXiv:1810.07366, 2018.
- [56] Wang A, Singh A, Michael J, et al. GLUE: A multi-task benchmark and analysis platform for natural language understanding[J]. arXiv preprint arXiv:1804.07461, 2018.
- [57] Yelp Dataset. Yelp Inc., [Year]. Available: <https://www.yelp.com/dataset>
- [58] Maas A, Daly R E, Pham P T, et al. Learning word vectors for sentiment analysis[C]//Proceedings of the 49th annual meeting of the association for computational linguistics: Human language technologies. 2011: 142-150.
- [59] Jack FitzGerald, Christopher Hench, Charith Peris, Scott Mackie, Kay Rottmann, Ana Sanchez, Aaron Nash, Liam Urbach, Vishesh Kakarala, Richa Singh, Swetha Ranganath, Laurie Crist, Misha Britan, Wouter Leeuwis, Gokhan Tur, and Prem Natarajan. 2022. Massive: A 1m-example multilingual natural language understanding dataset with 51 typologically-diverse languages.
- [60] Eneko Agirre, Daniel Cer, Mona Diab, and Aitor Gonzalez-Agirre. 2012. Semeval-2012 task 6: A pilot on semantic textual similarity. In * SEM 2012: The First Joint Conference on Lexical and Computational Semantics–Volume 1: Proceedings of the main conference and the shared task, and Volume 2: Proceedings of the Sixth International Workshop on Semantic Evaluation (SemEval 2012), pages 385–393.
- [61] Liu, Xin, Qingcai Chen, Chong Deng, Huajun Zeng, Jing Chen, Dongfang Li, and Buzhou Tang. "Lcqmc: A large-scale chinese question matching corpus." In Proceedings of the 27th international conference on computational linguistics, pp. 1952-1962. 2018.
- [62] Yang, Yinfei, Yuan Zhang, Chris Tar, and Jason Baldridge. "PAWS-X: A cross-lingual adversarial dataset for paraphrase identification." arXiv preprint arXiv:1908.11828 (2019).
- [63] Cer, Daniel, Mona Diab, Eneko Agirre, Inigo Lopez-Gazpio, and Lucia Specia. "Semeval-2017 task 1: Semantic textual similarity-multilingual and cross-lingual focused evaluation." arXiv preprint arXiv:1708.00055 (2017).
- [64] Tri Nguyen, Mir Rosenberg, Xia Song, Jianfeng Gao, Saurabh Tiwary, Rangan Majumder, and Li Deng. 2016. MS MARCO: A human generated machine reading comprehension dataset. In Proceedings of the Workshop on Cognitive Computation: Integrating neural and symbolic approaches 2016 co-located with the 30th Annual Conference on Neural Information Processing Systems (NIPS 2016), Barcelona, Spain, December 9, 2016, volume 1773 of CEUR Workshop Proceedings. CEUR-WS.org.

- [65] Tom Kwiatkowski, Jennimaria Palomaki, Olivia Redfield, Michael Collins, Ankur Parikh, Chris Alberti, Danielle Epstein, Illia Polosukhin, Jacob Devlin, Kenton Lee, et al. Natural questions: a benchmark for question answering research. *Transactions of the Association for Computational Linguistics*, 7:453–466, 2019.
- [66] Angela Fan, Yacine Jernite, Ethan Perez, David Grangier, Jason Weston, and Michael Auli. 2019. ELI5: Long Form Question Answering. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 3558–3567, Florence, Italy. Association for Computational Linguistics.
- [67] Zhilin Yang, Peng Qi, Saizheng Zhang, Yoshua Bengio, William Cohen, Ruslan Salakhutdinov, and Christopher D. Manning. HotpotQA: A dataset for diverse, explainable multi-hop question answering. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, pp. 2369–2380, Brussels, Belgium, October-November 2018. Association for Computational Linguistics. doi: 10.18653/v1/D18-1259. URL <https://aclanthology.org/D18-1259>.
- [68] Xinyu Zhang, Nandan Thakur, Odunayo Ogundepo, Ehsan Kamaloo, David Alfonso-Hermelo, Xiaoguang Li, Qun Liu, Mehdi Rezagholizadeh, and Jimmy Lin. Miracl: A multilingual retrieval dataset covering 18 diverse languages. *Transactions of the Association for Computational Linguistics*, 11:1114–1131, 2023.
- [69] Pranav Rajpurkar, Jian Zhang, Konstantin Lopyrev, and Percy Liang. Squad: 100,000+ questions for machine comprehension of text. *arXiv preprint arXiv:1606.05250*, 2016.
- [70] James Thorne, Andreas Vlachos, Christos Christodoulopoulos, and Arpit Mittal. Fever: a large-scale dataset for fact extraction and verification. *arXiv preprint arXiv:1803.05355*, 2018.
- [71] Wei He, Kai Liu, Jing Liu, Yajuan Lyu, Shiqi Zhao, Xinyan Xiao, Yuan Liu, Yizhong Wang, Hua Wu, Qiaoqiao She, Xuan Liu, Tian Wu, and Haifeng Wang. 2018. DuReader: a Chinese Machine Reading Comprehension Dataset from Real-world Applications. In *Proceedings of the Workshop on Machine Reading for Question Answering*, pages 37–46, Melbourne, Australia. Association for Computational Linguistics.
- [72] Yichen Jiang, Shikha Bordia, Zheng Zhong, Charles Dognin, Maneesh Singh, and Mohit Bansal. 2020. HoVer: A Dataset for Many-Hop Fact Extraction And Claim Verification. In *Findings of the Association for Computational Linguistics: EMNLP 2020*, pages 3441–3460, Online. Association for Computational Linguistics.
- [73] Zhang X, Ma X, Shi P, et al. Mr. TyDi: A multi-lingual benchmark for dense retrieval[J]. *arXiv preprint arXiv:2108.08787*, 2021.
- [74] Kyle Lo, Lucy Lu Wang, Mark Neumann, Rodney Kinney, and Daniel Weld. 2020. S2ORC: The Semantic Scholar Open Research Corpus. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 4969–4983, Online. Association for Computational Linguistics.

[75] <https://huggingface.co/spaces/mteb/leaderboard>

[76] Jinhyuk Lee, Feiyang Chen, Sahil Dua, Daniel Cer, Madhuri Shanbhogue, Iftexhar Naim, Gustavo Hernandez ´ Abrego, Zhe Li, Kaifeng Chen, Henrique Schechter Vera, et al. Gemini embedding: Generalizable embeddings from gemini. arXiv preprint arXiv:2503.07891, 2025b.

A Appendix

A.1 Framework Constraints

Table 4: Specifications of framework constraints

Item	Explanation
Keep core semantics	Preserving the core semantic content, which is the most critical requirement.
Diversity in morphology, syntax, grammar, tense, rhetoric, etc	Variations in lexical composition, syntactic structure, grammatical rules, and tense usage are permitted.
Length within $\pm 15\%$	The length deviation from the original sentence should not exceed 15%.
Keep language	The language used must be consistent with the original sentence.
Close in field	The content must remain strictly aligned with the domain of the given sentence.
Topic transfer, expansion, extension, prohibiting pure rewriting	Topic shifting, extension, or elaboration is permitted, but purely paraphrased content (identical to the original topic) is prohibited.
POS is the perfect answer (necessary & sufficient)	Positive examples must be unambiguous and precisely address the query (necessity condition) while containing exclusively relevant content without extraneous information (sufficiency condition).
Hard NEG: Worse than POS: - Semantic deviation (inadequate) - Including irrelevant information (unnecessary) - Different aspects of the same topic	Hard negative examples must exhibit inferior quality compared to positive instances, with noise introduced through three strategies: 1) semantic deviation (failing to accurately address the query), 2) incorporation of irrelevant information, or 3) maintaining the same topic but diverging in aspects.
Imitation: syntax, sentence structure, structural	Generating hard negative examples by emulating the structural and syntactic patterns of the given positive instance is a critical step to maximize discriminative challenge for the model.

A.2 Instruction Examples

Table 5: Instruction for partial training data

Dataset	Instruction
Huatuo	Given a medical question, retrieve user replies that best answer the question
Reddit	Retrieve the paragraph most semantically similar to the given statement
Law-GPT	Retrieve relevant legal provisions or interpretations for the given case
MNLI/SNLI	Retrieve semantically similar text
Yelp	Classify the customer review of businesses
Weibo	Classify the sentiment of Weibo comments

A.3 Data Synthesis Examples

Note: The text highlighted in yellow represents the original sentence, followed by the synthetically generated sentence.

Table 6: Paraphrasing Example (1)

query	pos
What is the best credit card for someone with no credit history?	If you’ve never had a credit card before a likely reason can be due to lack of credit history. You can apply for a department store card.
What’s the ideal credit card for a person without any credit history?	If you’ve never had a credit card, it’s probably because you don’t have a credit history. A department store card could be a good option to apply for.
What’s the top credit card choice for someone who has no credit history?	If you’ve never owned a credit card, it’s probably because you don’t have a credit history. A department store card might be a good option to consider.

Table 7: Paraphrasing Example (2)

query	pos
Which English Poet Laureate wrote 'The Faerie Queene'?	English Renaissance to begin, shakily, in the 1520s, and it continued until perhaps 1620. England had a strong tradition of literature in the English vernacular, which gradually increased as English use of the printing press became common during the mid 16th century. By the time of Elizabethan literature a vigorous literary culture in both drama and poetry included poets such as Edmund Spenser, whose verse epic 'The Faerie Queene' had a strong influence on English literature but was eventually overshadowed by the lyrics of William Shakespeare, Thomas Wyatt and others. Typically, the works of these playwrights and poets circulated in manuscript form.
Who was the English Poet Laureate that penned 'The Faerie Queene'?	The English Renaissance began, albeit slowly, in the 1520s and likely lasted until around 1620. England boasted a robust literary tradition in the English language, which expanded as the printing press became more widely adopted during the mid-16th century. By the time of Elizabethan literature, a dynamic literary scene in both drama and poetry emerged, featuring writers like Edmund Spenser, whose epic poem The Faerie Queene significantly influenced English literature, although it was eventually overshadowed by the works of William Shakespeare, Thomas Wyatt, and others. Generally, the writings of these poets and playwrights circulated in manuscript form.
Which English Poet Laureate authored 'The Faerie Queene'?	The English Renaissance began somewhat uncertainly in the 1520s, continuing through to around 1620. England had a rich literary heritage in its native tongue, which expanded as the printing press became more widely used in the mid-16th century. By the Elizabethan era, a thriving literary culture in both drama and poetry emerged, with poets like Edmund Spenser, whose epic The Faerie Queene profoundly impacted English literature, though it was eventually eclipsed by the works of William Shakespeare, Thomas Wyatt, and others. Generally, the writings of these playwrights and poets were circulated in manuscript form.

Table 8: Augmentation Example

query	pos	neg
What is the best credit card for someone with no credit history?	If you’ve never had a credit card before a likely reason can be due to lack of credit history. You can apply for a department store card.	-
Which credit card is easiest to get approved for with bad credit?	For those with poor credit scores, secured credit cards from Capital One or Discover are often the most accessible options since they require a refundable deposit.	Some premium travel cards like Chase Sapphire have great rewards but usually need excellent credit. Store cards may accept lower scores but have high interest rates.
What credit cards can I get as a college student with no credit?	Students without credit history often qualify for starter cards like Discover it Student or Capital One Journey, which are designed to help build credit.	Premium cards like American Express Platinum require good credit history, though some banks offer student accounts with debit cards.
Which English Poet Laureate wrote 'The Faerie Queene'?	...By the time of Elizabethan literature a vigorous literary culture in both drama and poetry included poets such as Edmund Spenser, whose verse epic 'The Faerie Queene' had a strong influence on English literature but was eventually overshadowed by the lyrics of William ...	-
What major epic poem did Edmund Spenser write during Queen Elizabeth's reign?	Edmund Spenser composed 'The Faerie Queene', an allegorical epic poem that became one of the most significant works of Elizabethan literature though later eclipsed by Shakespeare's popularity.	Christopher Marlowe's 'Hero and Leander' was another notable Elizabethan poem, but unlike Spenser's work it wasn't an epic allegory.
Which poet created 'Paradise Lost' during the English Renaissance?	John Milton authored the epic poem 'Paradise Lost' in the 17th century, a monumental work that explored biblical themes through blank verse and became a cornerstone of English literature.	William Blake's 'The Marriage of Heaven and Hell' also dealt with religious themes, though it was more prophetic than epic in style compared to Milton's masterpiece.

Table 9: Hard-Negative Generation Example

query	pos	neg
What territory was King Hussein afraid Israel would obtain?	...Hussein was nonetheless wary that an Egyptian-Israeli war would risk the West Bank's occupation by Israel...	-
What territory was King Hussein afraid Israel would obtain?	...Hussein was nonetheless wary that an Egyptian-Israeli war would risk the West Bank's occupation by Israel...	King Hussein expressed concerns about potential Israeli expansion during the Arab-Israeli conflicts, though his warnings to Nasser were delayed and initially dismissed, while other Arab leaders focused more on direct military preparations against Israel.
What territory was King Hussein afraid Israel would obtain?	...Hussein was nonetheless wary that an Egyptian-Israeli war would risk the West Bank's occupation by Israel...	King Hussein expressed concerns about potential Israeli territorial expansion during the 1967 tensions, though his warnings were delayed in reaching Nasser and mixed with broader regional tensions, while Egyptian military movements in Sinai were already underway under Amer's orders.